

Rajatsubhra Chakraborty

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SUMMARY

PhD candidate in Computer Science focused on intersection of bias, fairness and interpretability in generative models. Expertise in computer vision, deep learning, multimodal LLM and ethical AI deployment. Seeking internship opportunities to apply and explore ideas around vision language models, bias mitigation in generative models and interpretable AI systems.

EDUCATION

University of North Carolina, Charlotte

Aug 2023 – Dec 2027

PhD in Computer Science — GPA: 3.7/4.0

- Advisor: **Dr. Depeng Xu** and **Dr. Cori Faklaris**
- Research Areas: **Bias Mitigation and Discovery in Stable Diffusion**

Future Institute of Engineering and Management, Kolkata

Aug 2017 – Aug 2021

B.Tech in Computer Science — DGPA: 8.88/10

- Advisor: **Dr. Shibaprasad Sen**
- **Top 5%** of class
- Thesis: **Online Handwritten Bengali and Devanagari Character Recognition**

EXPERIENCE

PhD Pre-Candidate Researcher | [Trustworthy AI Lab, UNC Charlotte](#)

Jan 2025 – Present

- Proposed **BiasMap**, a model-agnostic pipeline that **localizes and quantifies entanglement** between demographic and semantic concepts in Stable Diffusion via cross-attention attribution; introduced **IoU/mBioU** as spatial bias metrics beyond output parity. [5]
- Designed **energy-guided diffusion** that minimizes a differentiable **SoftIoU** objective during denoising to **reduce concept overlap without retraining**, enabling plug-and-play mitigation.[1]
- *Ongoing*: researching **causal models for bias mitigation** in diffusion/T2I.

Graduate Research Assistant | [CharML Lab, UNC Charlotte](#)

Aug 2023 - Jan 2025

- Contributed to the development of **LLAVIDAL**, a large language vision model (LLVM) designed to understand daily activities of living (ADLs) by integrating video, 3D poses, and object trajectories. [6]
- Curated the **ADL-X** dataset, comprising of 100K video-instruction pairs enriched with 3D skeletons and object trajectories for multimodal training of LLMs.
- Developed a semi-automated framework for generating **pose-driven video descriptions** to enhance model alignment with human activities in ADL contexts.
- Benchmarked LLAVIDAL against SOTA models, achieving **+44% improvement** in action forecasting tasks.
- Engineered a curriculum-based **semi-supervised object detection** approach, achieving a **32% increase** in mAP for aerial object detection using 20% ground images. [8]
- Enhanced multimodal learning techniques by incorporating **geography-aware models**, improving aerial visual perception across geographies.

Student Research Assistant | [PURE Lab, Wayne State University](#)

Mar 2022 - Jul 2022

- Engineered a **U-Net-based deep learning model for cervical texture segmentation** in ultrasound images, improving segmentation accuracy and clinical usability.
- Developed an **automatic segmentation pipeline** for medical imaging by implementing **multiclass segmentation**, classifying regions into distinct tissue types.

Research Intern | [Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany](#)

Oct 2020 - Dec 2020

- Designed **deep local descriptors** for instance-level recognition of historical architecture using CNNs.

Research Intern | [CMATER Lab, Jadavpur University, Kolkata, India](#)

Jul 2020 - Aug 2020

- Developed transfer learning models for handwritten character recognition, achieving **99.99% accuracy** using the VGG16 architecture. [10, 11]

Research Intern | [Indian Statistical Institute Kolkata](#)

May 2020 - Jul 2020

- Designed automatic text detection pipeline for Indus Valley Civilization Seals.

PUBLICATIONS

(C) Conference, (W) Workshop, (J) Journal, (P) Preprint

1. (P) Chakraborty R., **Chakraborty R.**, Palaiahnakote S., Pal U., Cangelosi A. (2026). "Vision-Language Models are Fragile Multilingual Associators." [Under review](#)
2. (J) **Chakraborty R.**, Xu D., Faklaris C., Niu X., Wu X. "A Survey on Bias Discovery and Mitigation in Stable Diffusion Models." [Under review](#), 2026.
3. (C) **Chakraborty R.**, Espinosa-Momox A., Haskin R., Xu D., Porras-Aguilar R. "DM-QPMNET: Dual-modality fusion network for cell segmentation in quantitative phase microscopy." **IEEE International Conference on Image Processing (ICIP)**, 2026. [\[Paper\]](#)
4. (C) **Chakraborty R.**, Che X., Xu D., Faklaris C., Niu X., Yuan S. "BiasMap: Leveraging Cross-Attentions to Discover and Mitigate Hidden Social Biases in Text-to-Image Generation." **ACM KDD 2026**, 2026. [\[Paper\]](#)
5. (W) Chakraborty R., **Chakraborty R.**, Dasgupta A., Chaurasia S. "Do We Need Large VLMs for Spotting Soccer Actions?" **Proceedings of IJCNLP-AAACL Student Research Workshop (Oral)**, 2025. [\[Paper\]](#)
6. (W) Chakraborty R., **Chakraborty R.**, Rahimian A. K. "TruthLens: A Training-Free Paradigm for DeepFake Detection." **ICML DIGBUGS Workshop**, 2025. [\[Paper\]](#)
7. (W) **Chakraborty R.**, Che X., Xu D., Faklaris C., Niu X., Yuan S. "BIASMAP: Can Cross-Attention Uncover Hidden Social Biases?" **CVPR DemoDiv Workshop**, 2025. [\[Paper\]](#)
8. (C) Reilly D., **Chakraborty R.**, Sinha A., Govind M., Wang P., Bremond F., Xue L., Das S. "LLAVIDAL: Benchmarking Large Language Vision Models for Daily Activities of Living." **CVPR**, 2025. [\[Paper\]](#) [\[Website\]](#)
9. (W) Reilly D., **Chakraborty R.**, Sinha A., Govind M., Wang P., Bremond F., Xue L., Das S. "LLAVIDAL: A Large Language Vision Model for Daily Activities of Living." **NeurIPS MARS Workshop**, 2024. [\[Paper\]](#)
10. (C) Dutta A., Das S., Nielsen J., **Chakraborty R.**, Shah M. "Multiview Aerial Visual Recognition (MAVREC): Can Multi-view Improve Aerial Visual Perception?" **CVPR**, 2024. [\[Paper\]](#) [\[Website\]](#)
11. (J) Bhattacharyya A., **Chakraborty R.**, Saha S., Sen S., Sarkar R. "A Two-Stage Deep Feature Selection Method for Online Handwritten Bangla and Devanagari Character Recognition." **SN Computer Science**, 2022. [\[Paper\]](#)
12. (C) **Chakraborty R.**, Mukherjee D., Bhattacharyya A., Mukherjee H., Sur M. K., Sen S., Roy K. "Online Handwritten Bangla and Devanagari Character Recognition by Using CNN: A Deep Learning Concept." **IEEE ICCE**, 2020. [\[Paper\]](#)
13. (C) **Chakraborty R.**, Saha S., Bhattacharyya A., Sen S., Sarkar R., Roy K. "Recognition of Online Handwritten Bangla and Devanagari Basic Characters: A Transfer Learning Approach." **IAPR CVIP**, 2020. [\[Paper\]](#)
14. (J) Rudra M., Reddy P. S., **Chakraborty R.**, Sarkar P. P. "Design of Frequency Selective Surface Comprising of Dipoles Using Artificial Neural Network." **International Journal of Advances in Applied Sciences**, 2020. [\[Paper\]](#)

TEACHING EXPERIENCE

Lead Graduate Teaching Assistant | *UNC Charlotte*

Jan 2025 - Present

- Lead TA for **ITSC 3146 – Computer Networks and Operating Systems**.
- Designed course materials, lab assignments, and assessments for the course.
- Coordinated between instructors and TAs to ensure smooth course delivery.
- Managed and supervised all lab sessions, providing hands-on guidance to students.

Graduate Teaching Assistant | *Wayne State University, Detroit*

Aug 2022 - May 2023

- Taught and assisted in both **undergraduate and graduate-level** courses.
- Handled lectures, lab sessions, and grading for: **CSC 1100 - Problem Solving and Programming, CSC 2110 - Computer Science 1, CSC 7760 - Deep Learning, CSC 5750 - Principles of Web Technologies**.
- Mentored students in **Python, C++, and machine learning frameworks**, providing additional support for coursework and research.

TECHNICAL SKILLS

Languages: Python, C++, C

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Pandas

Domains: Computer Vision, Fairness in AI, Bias Mitigation, Multimodal LLM, Large Language Vision Models, Deep Learning, Image & Video Processing

Soft Skills: Research, Technical Writing, Communication, Collaboration

ACADEMIC SERVICES

Program Committee Member: PAKDD 2026, ICML DIGBUGS Workshop 2025

Reviewer: CVPR ReGenAI Workshop 2025, CVPR DemoDiv Workshop 2025, ISMB 2024, IEEE TMM 2025, ACL SRW 2025, CIKM 2025, EMNLP 2025, IEEE Transactions in Multimedia

STUDENT MENTORING

Monica Phann , Undergraduate Student, UNC Charlotte <i>Research Focus: Bias Mitigation in Diffusion Models and VLMs</i>	<i>Fall 2025 – Present</i>
Kevin Richards , Graduate Student, UNC Charlotte <i>Research Focus: Multimodal LLMs in UAV LiDAR</i>	<i>Fall 2025 – Present</i>